Abhin Suresh

Computational Physicist · Software Engineer · Doctoral Dissertation Award Recipient (2022-2023)
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Professional Summary

• Computational physicist with extensive (5+ years) experience in quantum transport device simulation, proven by **9 peer-reviewed publications** and a **doctoral dissertation award**.
• Skilled in open quantum systems, DFT Simulations [6], ultrafast dynamics, and quantum-classical hybrid simulations with a robust foundation in **nonequilibrium many-body physics**.
• Quantified long-range dynamical entanglement in spin valve spintronic device and proposed an experimental method to witness entanglement. [8]
• Developed computational models to analyze terahertz emission from magnetic materials [7] and to provide a detailed description of many-body excited states in spintronic system [4,5].
• Proficient in **Python** and efficient with **C++, MATLAB, UNIX**, and high-performance computing.

Education

Ph.D.: Department of Physics and Astronomy, University of Delaware. *2018 - 2025*

- **Thesis:** Ultrafast-light- or current-driven classical and quantum nonequilibrium states of magnons in magnetic heterostructures.
- **GPA:** 3.719/4
- **Award:** *University Dissertation Fellowship Award - University of Delaware.*
- **Publication:** 8 publications in peer-reviewed journals - Abhin Suresh Google Scholar page.

Skills

Physics	Quantum transport device simulation (multiscale, quantum-classical hybrid), Nonequilibrium many-body physics, Quantum Statistical Mechanics.
Experimental	Nano-composite Fabrication, Scanning Electron Microscopy, Powder-XRD, Raman Spectroscopy, Chemical Vapor Deposition, Thin Film Deposition.
Programming	Python, MATLAB, Fortran, Julia, C++, SQL, ngspice (SPICE Simulations) Bash scripting, GitHub version-control, Parallel programming-OpenMP/MPI, CUDA.
Numerical Methods	Low-rank Truncation, Atomistic Spin Simulation (LLG), Electromagnetic Simulation, Density Functional Theory, Density Matrix Renormalization Group, Tensor Network.
Machine Learning	Random Forrest Classifier/Regression, Gradient Boosting, Agglomerative Clustering, Reinforcement Learning, Neural Network, Data Wrangling, PyTorch, Scikit-Learn.

Experience

University of Delaware | Quantum Transport, Quantitative Data Analysis
Research Assistant | Newark 08/2019 - present

❖ Numerical simulation of electrons interacting with magnetic moments. [PUB: 2-3]
Quantitative Analysis | Python | FORTRAN | Scikit-Learn

- **Developed a quantum-classical** hybrid method to study self-consistent interaction of **quantum electrons** with magnetic moments using time-dependent non-equilibrium Green-function (TDNEGF) theory coupled to Landau-Lifshitz-Gilbert equation.
- **Implemented TDNEGF** theory (an integro-differential equation) as a set of linear differential equations to solve a time-dependent problem that scales linearly with time.
- **Resulted** in understanding the source of pumped charge and spin currents from precessing magnetic materials and how such materials can transmit spin currents.
- Implemented **Neural Networks** to predict the error during the time evolution of a truncated wavefunction for 2D TDNEGF.

IISER Thiruvananthapuram | Nano-Composite Fabrication and Characterization
Research Intern | Thiruvananthapuram 08/2014 - 08/2015

- ❖ Synthesis and characterization of Li-ion energy storage devices. [PUB: 1]
  Battery cycle analysis | Nanomaterial Fabrication and characterization
  - Synthesized metal-oxide nanocomposites ($TiNb_2O_7/Nb_2O_5$ with graphene) for electrochemical energy storage device (Li-ion battery and supercapacitors).
  - Material characterization using Powder XRD, Raman Spectroscopy, SEM, and TEM.
  - Device performance analysis using charge-discharge cycle.

University of Delaware| Machine Learning, Data Wrangling, Reinforcement Learning
Graduate Coursework Projects | Newark 03/2020 - 08/2020

- ❖ Comprehensive study on Time Series Analysis and Forecasting through research projects.
  Behavior Prediction | Gradient Boosting | Signal Processing | t-distributed Stochastic Neighbor Embedding | Agglomerative Clustering | Facebook Prophet | Autoregressive-MA
  - Voice Recognition sound bite analysis, LIGO data analytics.
  - Stock Market correlation and forecasting; Earthquake prediction.
  - Uber rides seasonal trend analysis and model validation.
- ❖ Study on implications of diversity in corporate leadership on economic performance.
  Random Forrest | Statistical Modelling | Neural Network | ROC
  - Developed and implemented Random forest classifier/regression models to explore the impacts of gender discrepancies.
  - Implemented Data collection, integration, and preprocessing; Imputation via KNN algorithms and balancing the datasets using bootstrap methods; Statistical modeling to validate model performance.

University of Delaware | Nonequilibrium Many-Body Physics, Ab Initio Calculations
Research Assistant | Newark 08/2019 - present

- ❖ Multiscale modeling of antiferromagnetic Weyl semimetal Mn_3Sn. [PUB: 7]
  Python | Julia | OpenMP/MPI | Cross-functional Collaboration
  - Developed quantum-classical hybrid framework combined with **Maxwell's equations** to study dynamics of magnetic materials excited by an ultrafast light pulse.
  - Designed numerical model to substantiate the experimental observations.
  - Studied the Hamiltonian-based microscopic mechanism behind terahertz and high-harmonic emission from magnetic heterostructures.
  - Implemented **multithreading** to efficiently scan emitted radiation on the surface.
- ❖ Numerical package development to study quantum many-body effects. [PUB: 4, 5, 8]
  Software Development | Parallel Programming | Open Quantum System | MATLAB
  - Tested the **numerical validation** of treating quantum spins as classical vectors and the mapping spin-operators to the widely used Holstein-Primakoff transformation.
  - **Implemented MPI/OpenMP** with **1024 PEs** to efficiently time-evolve a quantum system attached to multiple thermal baths using the method of quantum trajectories.
  - Leveraged Sparsity of matrices, and conservation laws to optimize computational cost.
  - Studied **quantum spins** in the presence of a **dissipative environment**.
  - **Quantified long-range dynamical entanglement** in spin valve spintronic device and proposed experimental method to witness the entanglement.
- ❖ Extracting spin mixing conductance parameter from ab initio calculations. [PUB: 6]
  Quantum ATK | Regression Analysis | Floquet-NEGF
  - Numerical simulation of spin pumping from antiferromagnetic insulators using **ab initio DFT study** and **validation** of results with TDNEGF-LLG.

Volunteer Work

- **Science outreach** Led and took part in the team to design science illustration cards to communicate our center's research activities in MRSEC Science Slam (a five-minute long slam on a research highlight or unique broader impact accomplishment of each center). Feb 2022
- **Seminar Host** Picked a speaker that fits our center's research goals. Took part in inviting, scheduling, and organizing the seminar. Dec 2021
- **Science outreach** Organized science Olympiad experiments in a primary school. Nov 2021
- **Git workshop** Organized "How to use Git" workshop as a part of University of Delaware Graduate Student Government Hackathon. April 2021

Conference and Talks

- **Workshop** Parallel programming, XSEDE HPC Workshop on MPI and CUDA. May 2021
- **Contributed talks and posters** Four Research talks/poster at APS March Meeting
- **Workshop** Summer School on Quantum Magnetism (EPFL Switzerland). Aug. 2019

Publications

9. D.-Q. To, C. Y. Ameyaw, **A. Suresh**, S. Bhatt, *et al.*, Giant spin Nernst effect in a two-dimensional antiferromagnet due to magnetoelastic coupling-induced gaps and interband transitions between magnon-like bands, Phys. Rev. B **108**, 085435 (2023).
8. **A. Suresh**, R. D. Soares, P. Mondal, J. P. Santos Pires, J. M. Viana Parente Lopes, *et al*., Electron-mediated entanglement of two distant macroscopic ferromagnets within a nonequilibrium spintronic device, Phys. Rev. A **109**, 022414 (2024).
7. **A. Suresh** and B. K. Nikolić, Quantum-classical approach to spin and charge pumping and the ensuing radiation in THz spintronics: Example of ultrafast-light-driven Weyl antiferromagnet Mn_3Sn, Phys. Rev. B **107**, 174421 (2023).
6. K. Dolui, **A. Suresh**, B. K. Nikolić, Spin pumping from antiferromagnetic insulator spin-orbit-proximitized by adjacent heavy metal: A first-principles Floquet-nonequilibrium Green function study, J. Phys. Mater. **5**, 034002 (2022), **Focus Issue on Nonequilibrium properties of quantum materials**.
5. P. Mondal, **A. Suresh** and B. K. Nikolić, When can localized spins interacting with conduction electrons in ferro- or antiferromagnets be described classically via the Landau-Lifshitz equation: Transition from quantum many-body entangled to quantum-classical nonequilibrium states, Phys. Rev. B **104**, 214401 (2021).
4. U. Bajpai, **A. Suresh**, B. K. Nikolić, Quantum many-body states and Green functions of nonequilibrium electron-magnon systems: Localized spin operators vs. their mapping to Holstein-Primakoff bosons, Phys. Rev. B **104**, 184425 (2021), **Editor's suggestion**.
3. **A. Suresh**, M.D. Petrović, U. Bajpai, *et al.*, Magnon- versus electron-mediated spin-transfer torque exerted by spin current across an antiferromagnetic insulator to switch the magnetization of an adjacent ferromagnetic metal, Phys. Rev. Appl **15**, 034089 (2021).
2. **A. Suresh**, U. Bajpai, and B. K. Nikolić, Magnon-driven chiral charge and spin pumping and electron-magnon scattering from time-dependent quantum transport combined with classical atomistic spin dynamics, Phys. Rev. B **101**, 214412 (2020).
1. P. Arunkumar, A. G. Ashish, B. Babu, S. Sarang, **A. Suresh**, *et al.*, Nb_2O_5/graphene nanocomposites for electrochemical energy storage, RSC Adv. **5**, 59997 (2015).

References

- Prof. Branislav K. Nikolić (Department of Physics and Astronomy, University of Delaware)